Mar & Kuku Production Studio

Complete MVP Guide, Verified Architecture & Production Roadmap (Milestones 1–9) • Updated October 2026

Current System Status & Integrity:
• **Production Asset Library:** `279 / 279` files in `/Users/yonassayfu/Desktop/DoroNesh_2D_Animation/` verified strictly read-only and byte-identical.
• **Automated Test Suite:** `61 passed` across all components (`.venv/bin/pytest -q`).
• **Technical Proof (M0 CH01):** `mk-studio verify-ch01` passed with `exit_code=0` (30.22s audio master, 4s interactive hold, native 5-track Palmier project, 907-frame verified MP4).
• **End-to-End Pipeline (Steps 1–9):** Fully implemented and verified from 3-Agent script consultation to YouTube Studio upload staging.

1. Executive Summary & Capabilities

Mar & Kuku Production Studio is an autonomous, owner-supervised production hub designed for high-quality 2D preschool animation. The system orchestrates multiple AI agents (ChatGPT, Google Gemini, Manus) via a managed browser session, synthesizes broadcast-quality audio, compiles native **Palmier Pro** edit timelines, renders deterministic 1080p30 video with sub-pixel cutout animation, and stages ready-to-publish packages for YouTube Studio.

2. The 9-Step End-to-End Production Pipeline

Step	Name & Target	Verified Execution & Output	Status
1	ChatGPT Master Plan	Drafts episode educational arc, scene structure, character dialogue, and specialist consultation prompts.	COMPLETED
2	Gemini Reviewer	Critiques preschool comprehension, vocabulary pacing, and 4-second interactive stillness hold.	COMPLETED
3	Manus Specialist	Analyzes speech tempo (110–120 WPM), 12-frame counting pauses, and character animation cues.	COMPLETED
4	ChatGPT Synthesis	Harmonizes master plan, reviewer feedback, and specialist timing into approved script (`v001`).	COMPLETED
5	Follow-Up Correction	Applies owner natural-language notes (e.g. counting petals) yielding `v002` without re-running 1–4.	COMPLETED
6	Asset Generation & Intake	Generates high-res visual props, 48kHz stereo voiceover master, 1280x720 thumbnail, and manifests.	COMPLETED
7	Palmier Timeline Editing	Compiles 5 tracks, 20 clips, and 6 markers into native `.palmier` project bundle on disk.	COMPLETED
8	Final MP4 Export & QA	Renders 1080p30 900-frame MP4; verifies Bob RMS (28.39) and Hold Stillness RMS (0.57).	COMPLETED

Step	Name & Target	Verified Execution & Output	Status
9	YouTube Studio Staging	Navigates managed browser to YouTube Studio, verifies dashboard status, and packages upload metadata.	COMPLETED

3. Studio Web UI & Interactive Controls

The local web application runs at `http://127.0.0.1:8765` and provides real-time visibility into all pipeline operations:

- **One-Click Action Buttons:** Header triggers for *1-4: 3-Agent Script*, *6: Assets*, *7: Palmier*, *8: MP4 Export*, *9: YouTube*, *Full Pipeline (6-9)*, *Resume*, *Verify CH01*, *Inspect Mac*, and *Stop*.
- **Natural-Language Chat:** Type instructions such as *"Research the next Mar and Kuku episode, consult Gemini and Manus, and save the proposed script"* or *"Revise the script to add three purple flowers"*.
- **Live Managed Browser Feed & Owner Takeover:** Real-time snapshot of the active Chrome session with instant *Owner Takeover* to pause automation for manual 2FA or CAPTCHA solving, and *Resume Auto* to continue safely.
- **Cooperative Stop:** A global stop signal halts running browser navigation, CLI provider queries, Mac actions, and workflow stages without losing checkpointed progress in `output_manifest.json`.

4. Step-by-Step Operator Guide

Step A: Launch Studio Web Interface

```
# Start the Studio server on port 8765
.venv/bin/mk-studio studio --port 8765

# Open your browser and navigate to:
http://127.0.0.1:8765/
```

Step B: Run the 3-Agent Script Consultation

In the Studio Chat input box, type:

```
Research the next Mar and Kuku episode, consult Gemini and Manus, and save the proposed scri
```

The system coordinates ChatGPT (Master), Gemini (Reviewer), and Manus (Specialist) in your managed Chrome browser, saving `proposed_script_v001.md` and checkpointing each step.

Step C: Trigger Full Production Pipeline (Assets → Palmier → MP4 → YouTube)

Click the **Full Pipeline (6-9)** button or type:

```
Run full production pipeline from script to YouTube Studio
```

Studio automatically generates the voiceover and teaching props, compiles the `.palmier` project, renders the 1080p30 video, runs computer vision QA on frame stillness, and stages the upload package in YouTube Studio.

5. Architecture & File Locations

Component	Key Path	Purpose
Studio Server	`src/mk_studio/studio/server.py`	ASGI application, chat routing, action triggers, and live UI.
Workflow Engine	`src/mk_studio/studio/workflow.py`	Checkpoint manager, step idempotency, and state persistence.
Production Pipeline	`src/mk_studio/production/pipeline.py`	Asset synthesis, Palmier compilation, MP4 rendering, and YouTube package.
Browser Controller	`src/mk_studio/browser/controller.py`	Chrome CDP automation, session management, and YouTube staging.
State Database	`scratch/ch01-proof/state.db`	SQLite database persisting conversations, jobs, telemetry, and artifacts.
Episode Workspace	`scratch/ch01-proof/episodes/EP03_COUNTING_FLOWERS/`	Generated scripts, voice WAVs, visual props, .palmier project, and final MP4.